Pergamon
0305-0491(94)E0028-R

Comp. Biochem. Physiol. Vol. 108B, No. 3, pp. 295–301, 1994
Copyright © 1994 Elsevier Science Ltd
Printed in Great Britain. All rights reserved
0305-0491/94 $7.00 + 0.00

Enzyme activities of glycogen metabolism and mitochondrial characteristics in muscles of RN^- carrier pigs (*Sus scrofa domesticus*)

Marielle Estrade, S. Ayoub, A. Talmant and G. Monin

Station de Recherches sur la Viande, INRA, Theix 63122 CEYRAT, France

High glycogen content and abnormal mitochondria have been seen in muscles from RN^- carrier pigs in a previous work. Glycogen synthase, branching enzyme, phosphorylase and debranching enzyme activities, and mitochondrial characteristics were studied in normal and RN^- carrier pigs. Branching enzyme activity was higher ($P < 0.01$) and glycogen synthase activity tended to be higher in longissimus dorsi muscle from RN^- carrier pigs compared to normal pigs. There were no differences in the activities of either phosphorylase and debranching enzyme between both types of pigs. Citrate synthase activity and mitochondrial respiration were slightly higher in muscle from RN^- pigs compared to normal pigs. Glycogen content in muscle from RN^- pigs could result from the imbalance between anabolic and catabolic enzyme activities of glycogen metabolism. The higher specific activity in mitochondria of RN^- pigs muscle might be the compensatory effect of an abnormal glycolytic metabolism.

Key words: RN^- carrier pigs; Glycogen; Glycogen synthase; Muscle mitochondria; Citrate synthase; Debranching enzyme; Glycogen phosphorylase; Enzyme activity; Glycogen metabolism.

Comp. Biochem. Physiol. 108B, 295–301, 1994.

Introduction

The dominant RN^- allele of the RN^-/rn^+ gene increases the glycogen content of pig white muscles (Monin, 1989; Marinova *et al.*, 1992; Fernandez *et al.*, 1992). White muscle fibres from RN^- carrier pigs (referred to as RN^-) showed an excess of up to 70% of glycogen in both sarcoplasmic and lysosomal compartments as compared with rn^+rn^+ homozygous pigs (referred to as normal pigs) (Estrade *et al.*, 1993). Moreover, a large proportion of mitochondria in white fibres of RN^- pig muscle showed an abnormal ultrastructure. These mitochondria contained scattered particles of glycogen and disorganized cristae (Estrade *et al.*, 1992).

Glycogen metabolism is partly controlled by four key enzymes: glycogen synthase and 1,4-α-glucan: 1,4-α-glucan 6-α-D-(1,4-αD-glucano) transferase (branching enzyme) regarding synthesis, glycogen phosphorylase and amylo-α-(1–6)-D-glucosidase (debranching enzyme) regarding glycogenolysis. The first aim of this work was to study whether changes in the activities of one or several of these enzymes could be involved

Correspondence to: G. Monin, Station de Recherches sur la Viande, INRA, Theix 63122 CEYRAT, France.
Received 7 September 1993; accepted 15 February 1994.

in the increase of glycogen content in RN^- pig muscle. The second aim of this study was to test the functional activity of mitochondria from RN^- pigs in order to verify if the oxidative metabolism is affected by the RN^- gene.

Materials and Methods

Animals

RN^- and normal pigs were selected according to the glycolytic potential of their longissimus dorsi muscle measured from a shot biopsy (Fernandez *et al.*, 1992). Glycolytic potential is used as an estimator of the glycogen level in the live resting animal. Indeed, a small part of glycogen may be degraded during the lapse of time (a few tenths of seconds) between the taking and the freezing of the biopsy sample. According to Fernandez *et al.* (1992), RN^- pigs can be discriminated from the population used in this study on the basis of a glycolytic potential higher than 210 μmol/g fresh weight of muscle.

The glycolytic potential was determined on muscle biopsies from 25 pigs of an experimental herd including RN^- carrier pigs (Talmant *et al.*, 1989). Glycolytic potential was estimated according to the following formula:

$$\text{Glycolytic potential} = 2([\text{glycogen}] + [\text{glucose-6-phosphate}] + [\text{glucose}]) + [\text{lactic acid}]$$

The sum of glycogen, glucose and glucose-6-phosphate was determined using the method of Dalrymple and Hamm (1973); lactic acid was determined according to Bergmeyer (1974).

Five RN^- pigs with a glycolytic potential from 250 to 297 μmol/g fresh muscle and five normal pigs with a glycolytic potential from 118 to 166 μmol/g fresh muscle were retained for this study. The animals were slaughtered in pairs at around 100 kg live weight by electronarcosis for 2 sec and exsanguination in an experimental slaughterhouse, each pair including one RN^- pig and one normal pig. For each animal, within 5 min of electronarcosis, six samples of 2 g of longissimus dorsi muscle were freeze-clamped in liquid nitrogen and kept at -80°C for determination of enzyme activities. Determinations were made in triplicate. Two slices of muscle (1–2 cm thick) were dipped in liquid nitrogen for further mitochondrial content determination. About 30 g fresh muscle were taken for immediate preparation of mitochondria.

Enzyme assays

One gram of frozen muscle was homogenized in 20 ml buffer (63 mM glycylglycine, 500 mM saccharose, 125 mM EDTA, 125 mM NaF, pH 7.4) and enzyme activities were determined on the supernatant obtained after centrifugation of the homogenate (2500 *g*, 20 min).

All enzyme assays were performed spectrophotometrically at 30°C except for debranching enzyme.

The glycogen synthase assay was based on the technique of Passonneau and Rottenberg (1973). Glycogen synthase activity was measured in the absence (synthase I) or in the presence (synthase I + D) of 10 mM glucose-6-phosphate. The state of activation of glycogen synthase was expressed as the ratio between the I form and the I + D forms in percent.

The branching enzyme assay was based on the technique of Gibson *et al.* (1971). This technique uses the reversibility of phosphorylase activity. Branching enzyme activity was determined from the amount of inorganic phosphate (P_i) released from G-1-P.

Glycogen phosphorylase *a* was assayed according to the technique of Bergmeyer (1974) and phosphorylase $a + b$ by adding AMP to the reaction medium (2 mM final concentration). The state of activation of phosphorylase was expressed as the ratio between activities of forms *a* and $a + b$ in percent.

Citrate synthase activity was measured by the technique of Srere (1969) using Ellman reagent at 412 nm.

Debranching enzyme activity was assayed by using the reversibility of the debranching reaction. The activity was obtained from the amount of ^{14}C glucose incorporated into glycogen during 1 hr incubation at 30°C (Taylor *et al.*, 1975) as determined by the filter paper method of Thomas *et al.* (1968).

Enzyme activities (except for branching and debranching enzymes) were expressed

in kat/mg protein. The activity of the branching enzyme was expressed in nmol P_i/sec/mg protein. The activity of debranching enzyme was expressed in pmol ^{14}C glucose/sec/mg protein.

The protein content of the homogenate was assayed by the method of Bradford (1976) using bovine serum albumin (BSA) as standard.

Mitochondrial preparation

Mitochondrial fractions were prepared according to the method of Idell-Wenger *et al.* (1982), except that homogenizing buffer was made according to Herpin and Barré (1989). About 5 min after slaughter, 30 g of longissimus dorsi muscle were removed from the animals and quickly immersed in ice-cold homogenizing buffer (5 ml/g fresh muscle) (100 mM saccharose, 180 mM KCl, 50 mM Tris, 10 mM EDTA, 5 mM $MgCl_2$, 1 mM ATP-K, pH 7.4). Muscles were minced with scissors. The mixture was washed three times with the same buffer. The suspension was added to 1% (w/v) of Nagarse solution in the same buffer in order to obtain a final concentration of 0.4 mg/ml (2 mg Nagarse/g tissue). The suspension was homogenized for 3 × 10 sec using a Polytron at low speed. The homogenate was diluted with the homogenizing buffer to 20 ml/g fresh muscle and filtered through two layers of cheesecloth. The homogenate was centrifuged at 450 *g* for 2 min. The supernatant was filtered again and centrifuged at 8000 *g* for 10 min. The pellet containing the mitochondria was suspended in buffer (150 mM KCl, 10 mM MOPS, pH 7.4).

Aliquots were removed for determination of citrate synthase activity and protein. The protein concentration was estimated by the Biuret method (Gornall *et al.*, 1949).

Measurement of oxidative phosphorylation

Oxygen consumption by mitochondria was measured polarographically with a Clark oxygen electrode at 25°C. The reaction medium contained 75 mM saccharose, 30 mM KCl, 20 mM KH_2PO_4, 6 mM $MgCl_2$ and 200–250 μg protein in a final volume of 1.6 ml at pH 7.4. State III respiration was initiated by the addition of 0.25 mM of ADP. The Respiratory Control Index (RCI) and ADP/O ratios were determined in the presence of glutamate plus malate (6 mM final concentration) or succinate (6 mM final concentration) plus rotenone (6 μM). RCI was calculated by dividing the rate of oxygen consumption in state III (in the presence of ADP) by the rate in state IV (after exhaustion of ADP by phosphorylation) as described by Chance and Williams (1956).

Mitochondrial content

The mitochondrial content was assessed from frozen tissue by measuring activity of citrate synthase in both isolated mitochondria and muscle homogenate and then applying the following equation:

$$\frac{\text{mg mitochondrial protein}}{\text{g muscle}} = \frac{\text{citrate synthase activity/g muscle}}{\text{citrate synthase activity/mg mitochondrial protein}}$$

Statistical analysis

Data were represented as mean ± SD. The mean values were compared by Student's *t*-test.

Results

Activities of glycogen synthesis enzymes in longissimus dorsi of RN⁻ and normal pigs muscle are reported in Fig. 1. Glycogen synthase activities (I and I + D forms) tended to be higher in the muscle from RN⁻ pigs as compared to normal pigs' muscle but the difference was not significant. The glycogen synthase I + D activity was 0.68 ± 0.1 in RN⁻ and 0.46 ± 0.2 nkat/mg protein in normal pigs' muscle. The glycogen synthase I/I + D ratio was not different between RN⁻ and normal pig muscle. On the other hand, branching enzyme activity was twice as high in the muscle of RN⁻ pigs than in the muscle of normal pigs ($P < 0.01$). The average values were 17.3 ± 2.5 and 8.0 ± 2.0 nmol/sec/mg protein, respectively.

The glycogen catabolic enzyme activities in RN⁻ and normal pigs' muscle are shown in Fig. 2. Both phosphorylase forms, the phosphorylase $a/a + b$ ratio and the debranching enzyme activities did not differ between RN⁻ and normal pigs' muscle. The

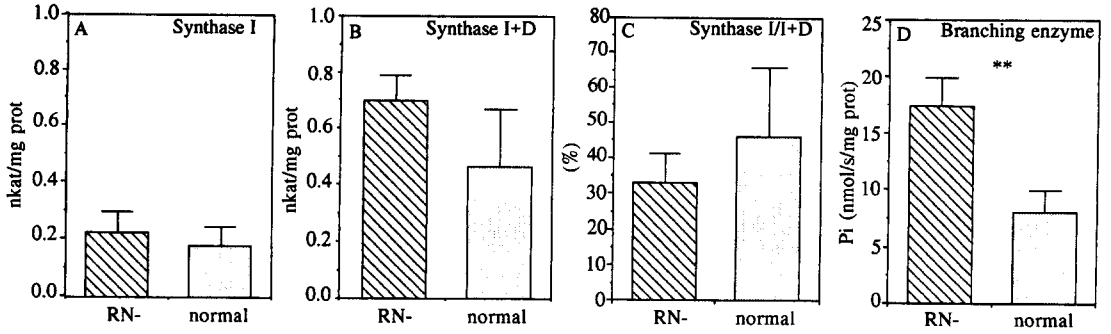


Fig. 1. Activities of glycogen synthase and branching enzyme in longissimus dorsi of RN$^-$ ($N = 5$) and normal ($N = 5$) pigs. Glycogen synthase (GS) activity was expressed in nkat/mg protein. A: Active GS form (synthase I) activity measured in the absence of G6P. B: Total form (synthase I + D) activity measured in the presence of 10 mM G6P. C: Degree of activation of GS (synthase I/I + D) expressed in %. D: Branching enzyme activity expressed in nmol of phosphate liberated/sec/mg protein. Values are means ± SD. Significant difference between RN$^-$ and normal pigs $^{**}P < 0.01$.

activity of phosphorylase $a + b$ was found to be 15 ± 2 and 16 ± 3 nkat/mg protein in RN$^-$ and normal pigs, respectively. The debranching enzyme activities were 4.3 ± 1.0 and 3.8 ± 0.6 nkat/mg protein in RN$^-$ and normal pig muscle, respectively.

Table 1 shows the mitochondrial characteristics. Citrate synthase activity was higher in longissimus dorsi muscle from RN$^-$ pigs than in longissimus dorsi muscle from normal pigs ($P < 0.01$).

State III oxygen consumption and RCI were higher in mitochondria from RN$^-$ pigs than in mitochondria from normal pigs in the presence of glutamate plus malate ($P < 0.05$). In the presence of succinate, no significant difference in RCI, state III and state IV oxygen consumption was observed. With both substrates there were no differences in ADP/O ratios between RN$^-$ and normal muscle mitochondria. Finally, the mitochondrial contents of longissimus dorsi from RN$^-$ and normal pigs were similar.

Discussion

No significant differences in activities of either glycogen synthase form were observed between RN$^-$ and normal pigs. This is in contrast to the relationship typically observed between glycogen content and glycogen synthase I + D activity in normal muscle, independently of muscle metabolic type. Under normal conditions, high levels of glycogen decrease activities of both glycogen synthase forms in skeletal muscle

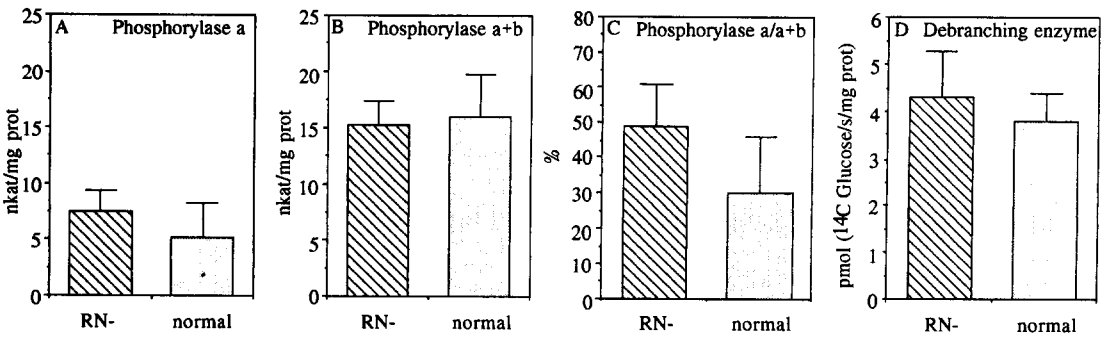


Fig. 2. Activities of phosphorylase and debranching enzymes in longissimus dorsi of RN$^-$ ($N = 5$) and normal ($N = 5$) pigs. Phosphorylase activity was expressed in nkat/mg protein. A: Phosphorylase a activity measured in the absence of AMP. B: Phosphorylase $a + b$ activity measured in the presence of 2 mM AMP. C: Degree of activation of phosphorylase $a/a + b$ expressed in %. D: Debranching enzyme activity expressed in pmol of ^{14}C glucose incorporated into glycogen/sec/mg protein. Values are means ± SD.

Table 1. Mitochondrial characteristics

		RN⁻		rn⁺
Citrate synthase activity	nkat/mg prot	1.0 ± 0.2	**	0.6 ± 0.1
Mitrochondrial content	mg mito/g meat	10.5 ± 3		11.5 ± 3
Mitochondrial respiration				
Glutamate plus malate	State IV	23.2 ± 4.6		25.2 ± 4.8
	State III	119.5 ± 3.3	*	98.8 ± 7.7
	RCI	6.3 ± 1.5	*	3.9 ± 0.7
	ADP/O	3.4 ± 0.4		3.3 ± 0.1
Succinate	State IV	30.9 ± 2.5		30.0 ± 5.5
	State III	112.3 ± 9.7		122.0 ± 34.0
	RCI	4.2 ± 1.1		3.6 ± 1.5
	ADP/O	1.9 ± 0.2		2.1 ± 0.1

Citrate synthase activity and mitochondrial content were obtained in mitochondria of 6 RN⁻ and 6 rn⁺ pigs muscle. Respiration was realized on mitochondria of RN⁻ ($N = 4$) and rn⁺ ($N = 2$) pigs. States III and IV oxygen consumptions are expressed in natoms O_2/min/mg mitochondrial proteins. Values are means ± SD. Significant difference between RN⁻ and rn⁺ *$P < 0.05$; **$P < 0.01$.

(Danforth, 1965; Villar-Pallasi, 1969; Munger *et al.*, 1993). Regarding branching enzyme activity, the present observations agreed with those of Bernard (1986). The latter author found a positive relationship between branching enzyme activity and muscle glycogen content. In pig longissimus dorsi muscles, both glycogen content and the branching enzyme activity were decreased concurrently by three times between 0 and 70 postnatal days. Nevertheless, in the majority of the reported cases, excessive amounts of glycogen in muscle have been associated with a deficiency or an apparent absence of the branching enzyme activity (Brown and Brown, 1968; Barash *et al.*, 1991). Such pathological states have been found in various tissue including muscle and liver. In these cases, the polysaccharide exhibits an abnormal structure (Brown and Brown, 1968; Mercier and Whelan, 1970). In the case of the RN⁻ gene, no abnormality was observed in the liver glycogen content (Monin *et al.*, 1992). Therefore, the relationship between the branching enzyme and glycogen content remains unclear. The glycogen branching enzyme catalyses the last step in glycogen biosynthesis by introducing α-1–6 branch points into the glycogen molecule. As noticed by Villa-Moruzzi *et al.* (1979), it may be that the metabolism of 1–6 glycosidic linkages plays some unknown role in the metabolism of glycogen. Thus, the red muscles, which usually contain lower stores of glycogen than the white muscles, show higher branching enzyme activity. Also, the length of the outer chains of glycogen are shorter in red than in white muscles (Dalrymple *et al.*, 1973). A modification in the activity of the branching enzyme in longissimus dorsi from RN⁻ pigs, compared with normal pigs, could induce an abnormal glycogen molecule structure with an excess of branch points.

The increases in branching enzyme, and the trend to a higher glycogen synthase activity observed in RN⁻ pigs could result from the higher oxidative activity characterizing the muscle of these pigs, as indicated by the higher citrate synthase activity and state III oxygen consumption. Activities of the glycogen synthase I + D forms and branching enzyme are higher in slow muscle than in fast muscle of the rabbit (Tsutou *et al.*, 1985) and of the rat (Villa-Moruzzi *et al.*, 1979). In pigs, Monin *et al.* (1987) have shown a similar relationship but only for the glycogen synthase I form.

Phosphorylase activities were slightly higher than those reported by Laborde *et al.* (1985) and Monin *et al.* (1986) in Hampshire crossbred pigs which could be considered as RN⁻ pigs (Monin, 1989). Electronarcosis by itself is likely to induce an increase in phosphorylase *a* activity as it results in pronounced muscle contraction. However, this occurs to the same extent in RN⁻ and rn⁺ pigs. Again, no relationship between glycogen phosphorylase activity and glycogen concentration was observed, although an increase in glycogen concentration has a negative effect on glycogen

synthase activity and a stimulatory effect on phosphorylase activity in human muscle (Munger *et al.*, 1993) or in rabbit liver (Mellgren and Coulson, 1983).

The second part of this study dealt with oxidative metabolism. A mitochondrial lesion was suspected in white fibres of RN^- pigs on the basis of morphological findings (Estrade *et al.*, 1992). In muscle cells, the presence of glycogen particles, or disorganized cristae in mitochondria, is generally considered as abnormal and has been reported to be associated with pathologies (Morgan-Hughes, 1982; Tassin and Brucher, 1982; Kamieniecka and Schmalbruch, 1980). Such ultrastructural abnormalities have been shown in several cases to be due to deficiencies in respiratory enzymes (Coquet, 1991). Sluga and Moser (1970) reported a metabolic abnormality which associated an excess in glycogen content and giant mitochondria with a lack of glycolysis stimulation.

The present results confirm earlier findings (Estrade *et al.*, 1992). The oxidative capacities tended to be higher in mitochondria from RN^- pigs, compared with normal pigs, since the RCI and state III oxygen consumption for oxidation of glutamate-plus-malate, and citrate-synthase activity were higher in longissimus dorsi muscle from RN^- pigs. This difference in citrate-synthase activity agrees with previous results of Monin *et al.* (1987). The present observations are surprising, since ultrastructural mitochondrial abnormalities are usually associated with decreases in respiratory enzymes. Nevertheless, two hypotheses can be put forward concerning the apparent discrepancy between ultrastructural and biochemical studies: (1) mitochondrial ultrastructural abnormalities were seen only in a part of the mitochondria of white fibres. Since the biochemical study was realized on a homogenate of muscle, it is possible that either the proportion of deficient mitochondria was not sufficient to affect significantly the values obtained from a muscle homogenate, or the abnormal mitochondria were eliminated during the preparation. (2) The amount of cristae, which are the support of the respiratory chains, is higher in most abnormal mitochondria than in normal ones (Estrade *et al.*, 1992). This abnormality could, therefore, be responsible for the higher measured activity.

Two main conclusions may be derived from the present observations:

(1) The increase in glycogen content characterizing the muscle of RN^- pigs could result from the imbalance between anabolic and catabolic enzyme activities of glycogen metabolism. The fact that the most important difference affects the branching enzyme activity suggests that the molecular structure of the glycogen molecules could be modified.

(2) The higher citrate synthase and respiratory activities in mitochondria of RN^- pig muscle might be a compensatory effect for an abnormal glycolytic metabolism. Indeed, considering that the oxidative and glycolytic metabolisms are markedly interdependent, there may also be a relationship between the high glycolytic potential and a disfunction of part of the mitochondria in RN^- white fibres.

References

Barash V., Lilling S., Fischer R. and Argov Z. (1991) Apparent absence of glycogen branching enzyme activity in phosphofructokinase deficiency. *J. Inher. Metab. Dis.* **14,** 902–907.

Bergmeyer H. N. (1974) In *Methods of Enzymatic Analysis*, 2nd edn, Vol. 1, pp. 505–507. Academic Press, New York.

Bernard D. (1986) Evolution des activités enzymatiques de la synthèse et de la dégradation du glycogène entre 0 et 70 jours dans le muscle de porc. *DEA Sc. Alimentaires.* Université de Clermont-Ferrand.

Bradford M. (1976) A rapid and sensitive method for quantification of microgram quantities of protein utilizing the principle of protein-dye binding. *Analyt. Biochem.* **72,** 248–254.

Brown B. I. and Brown D. H. (1968) Glycogen storage disease, 151–167.

Chance B. and Williams G. R. (1956) The respiratory chain and oxidative phosphorylation. In *Advances Enzymol.* **17,** 65–134.

Coquet M. (1991) Anomalies morphologiques des mitochondries dans les myopathie mitochondriales. In *Colloque Franco-Ibérique de Microscopie électronique.* 45 pp. Barcelona, Spain.

Dalrymple R. H. and Hamm R. (1973) A method for the extraction of glycogen and metabolites from a single muscle sample. *J. Food. Technol.* **8,** 439–444.

Dalrymple R. H., Kastenschmidt L. L. and Cassens R. G. (1973) Glycogen and phosphorylase in developing red and white muscle. *Growth* **37,** 19–34.

Danforth W. H. (1965) Glycogen synthetase activity in skeletal muscle. *J. biol. Chem.* **240,** 588–593.

Estrade M., Vignon X., Bazin C. and Monin G. (1992) Observation of abnormal mitochondria in white muscles fibres of RN⁻ carrier pigs. *38th ICoMST, Clermont-Ferrand, France* **3,** 351–354.

Estrade M., Vignon X., Rock E. and Monin G. (1993) Glycogen hyperaccumulation in white muscle fibres of RN⁻ carrier pigs—A biochemical and ultrastructural study. *Comp. Biochem. Physiol.* **104B,** 321–326.

Fernandez X., Tornberg E., Naveau J., Talmant A. and Monin G. (1992) Bimodal distribution of the muscle glycolytic potential in French and Swedish populations of Hampshire crossbred pigs. *J. Sci. Fd Agric.* **59,** 307–311.

Gibson B. W., Brown B. I. and Brown D. H. (1971) Studies of glycogen branching enzyme. Preparation and properties of α-1–4 glucan α-1–4 glucan 6-glycosyltransferase and its action on the characteristic polysaccharide of the liver of children with type IV glycogen storage disease. *Biochemistry* **23,** 4253–4255.

Gornall A. G., Baradwill C. J. and David M. M. (1949) Determination of serum proteins by means of buiret reaction. *J. biol. Chem.* **177,** 751–766.

Herpin P. and Barré H. (1989) Loose-coupled subsarcolemmal mitochondria from muscle Rhomboideus in cold-acclimated piglets. *Comp. Biochem. Physiol.* **92B,** 59–65.

Idell-Wenger J. A. I., Grotyohann L. W. and Neely J. R. (1982) An improved method for isolation of mitochondria in high yields from normal, ischemic and autolyzed rat hearts. *Analyt. Biochem.* **125,** 269–276.

Kamieniecka Z. and Schmalbruch H. (1980) Neuromuscular disorders with abnormal mitochondria. *Int. Rev. Cytol.* **65,** 321–357.

Laborde D., Talmant A. and Monin G. (1985) Activités enzymotiques metaboliques et contractiles de 30 muscle de porc. Relation avec le pH ultime alteint après le mort. *Reprod. Nut. Dévelop.* **25,** 619–628.

Marinova P., Lefaucheur L., Fernandez X. and Monin G. (1992) Relationship between metabolism and glycogen content in skeletal muscle fibers of Large White and Hampshire crossbred pigs. *J. Muscle Foods* **3,** 91–97.

Mellgren R. L. and Coulson M. (1983) Coordinated feed back regulation of muscle glycogen metabolism inhibition of purified phosphate by glycogen. *Biochem. biophys. Res. Commun.* **114,** 148–154.

Mercier C. and Whelan W. J. (1970) The fine structure of glycogen from type IV glycogen storage disease. *Eur. J. Biochem.* **16,** 579–583.

Monin G. (1989) Genetic effects on technological qualities of pig meat. *40th Ann. Meet. EAAP, Dublin* **1,** 3.

Monin G., Brard C., Vernin P. and Naveau J. (1992) Effects of the RN⁻ gene on some traits of muscle and liver in pigs. *38th ICoMST, Clermont-Ferrand, France* **3,** 391–394.

Monin G., Mejenes Quijano A. and Talmant A. (1987) Influence of breed and muscle metabolic type on muscle glycolytic potential and meat pH in pigs. *Meat Sci.* **20,** 149–158.

Monin G., Talmant A., Laborde D., Zabari M. and Sellier P. (1986) Compositional and enzymatic characteristics of the longissimus dorsi muscle from Large White, halothane-positive and halothane-negative Pietrain, and Hampshire pigs. *Meat Sci.* **16,** 307–316.

Morgan-Hughes J. A. (1982) Mitochondrial myopathies. In *Skeletal Muscle Pathology* (Edited by Mastaglia F. L. and Walton J.), Vol. 9, pp. 309–338.

Munger R., Temler E., Jallut D., Haesler E. and Felber J. P. (1993) Correlations of glycogen synthase and phosphorylase activities with glycogen concentration in human muscle biopsies—Evidence for a double-feedback mechanism regulating glycogen synthesis and breakdown. *Metab. Clin. Exp.* **42,** 36–43.

Passonneau J. V. and Rottenberg D. A. (1973) An assessment of methods for measurement of glycogen synthase activity including a new direct one step assay. *Analyt. Biochem.* **51,** 528–541.

Sluga E. and Moser K. (1970) Myopathy with glycogen storage and giant mitochondria (ultrastructural and biochemical findings) In *Muscle Diseases* (Edited by Walton J. N., Canal N. and Scarlato G.), pp. 116–119.

Srere P. A. (1969) Citrate synthase. In *Methods of Enzymatic Analysis* 2nd edn, Vol. 1, pp. 443–444. Academic Press, New York.

Talmant A., Fernandez X., Sellier P. and Monin G. (1989) Glycolytic potential in longissimus dorsi muscle of Large-White pigs, as measured after *in vivo* sampling. *35th ICoMST, 20–25 August, Copenhagen, Denmark* **3,** 1129–1132.

Tassin S. and Brucher J. M. (1982) The mitochondrial disorders: pathogenesis and aetiology classification. *Neuropathol. appl. Neurobiol.* **8,** 251–263.

Taylor C., Cos A. J., Kernohan J. C. and Cohen P. (1975) Debranching enzyme from rabbit skeletal muscle. *Eur. J. Biochem.* **512,** 105–115.

Thomas J. A., Schlender K. K. and Larner J. (1968) A rapid filter paper assay for UDP–glucose–glycogen glucosyltransferase, including an improved biosynthesis of UDP-C^{14} glucose. *Analyt. Biochem.* **25,** 486–499.

Tsutou A., Nakamura S., Negam A., Kobayashi T., Mizuta K., Hashimoto E. and Yamamura H. (1985) Comparison of enzyme activities on glycogen metabolism in rabbit slow and fast muscles. *Comp. Biochem. Physiol.* **81,** 641–645.

Villa-Moruzzi E., Locci-Cubeddu T. and Bergamini E. (1979) Developmental changes of glycogen enzymes in fast and slow muscles of the rat. *Growth* **43,** 73–79.

Villar-Palasi C. (1969) Oligo- and polysaccharide inhibition of muscle transferase D-phosphatase. *Ann. N.Y. Acad. Sci.* **166,** 719–730.